Hydrated & Anhydrous Salts

Exam code: 0620 & 0971

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Hydrated & anhydrous salts

- When salts are being prepared, some water can be retained within the structure of the salt during the crystallisation process
- Salts that contain water within their structure are called **hydrated** salts
- **Anhydrous** salts are those that contain no water in their structure
- A common example is copper(II) sulfate which crystallises forming the salt hydrated copper(II) sulfate, which is blue
- When it is heated, the water from its structure is removed, forming anhydrous copper(II) sulfate, which is white
- The hydrated salt has been **dehydrated** to form the anhydrous salt
- This reaction can be reversed by adding water to anhydrous copper(II) sulfate

hydrated copper(II) sulfate ⇌ anhydrous copper(II) sulfate + water

Water of crystallisation

Help me understand

Extended tier only

- Water molecules included in the structure of some salts during the crystallisation process are known as **water of crystallisation**
- A compound that contains water of crystallisation is called a **hydrated compound**
- When writing the chemical formula of hydrated compounds, the water of crystallisation is separated from the main formula by a **dot**:
  - Hydrated copper(II) sulfate is $CuSO_4 \cdot 5H_2O$
  - Hydrated cobalt(II) chloride is $CoCl_2 \cdot 6H_2O$
- The formula shows the number of moles of water contained within one mole of the hydrated salt:
  - Hydrated copper(II) sulfate, $CuSO_4 \cdot 5H_2O$, contains 5 moles of water in 1 mole of hydrated salt
- A compound which doesn't contain water of crystallisation is called an **anhydrous compound**:
  - Anhydrous copper(II) sulfate is $CuSO_4$
  - Anhydrous cobalt(II) chloride is $CoCl_2$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- $CuSO_4 + 5H_2O \rightarrow CuSO_4 \cdot 5H_2O$

- Hydrated to anhydrous salt (by heating):
  - $CuSO_4 \cdot 5H_2O \rightarrow CuSO_4 + 5H_2O$

Hydrated copper(II) sulfate and anhydrous copper(II) sulfate

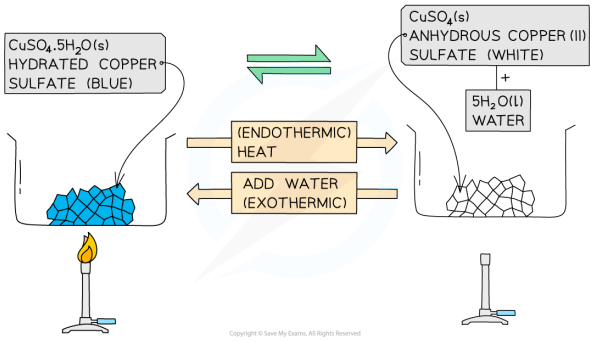

Diagram showing the dehydration of hydrated copper(II) sulfate

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